

NADTHAWUTH KETNUKKUNN SU PIANG

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PERSONAL PROFILE

Computer Science student at the University of Southampton with extensive experience architecting full-stack web applications, multi-tenant B2B SaaS platforms, and cloud-native systems. Proficient in modern TypeScript/Next.js ecosystems, enterprise Java/Spring Boot, and secure database design with PostgreSQL and Supabase. Passionate about software engineering and security-first architecture, with an active drive to expand into cybersecurity and low-level system defense.

EDUCATION

University Of Southampton, Southampton, UK

Sept 2024 - Present

BSc. Computer Science

Modules: Algorithmics, Computer Systems, Data Management, Mathematics, Networks and Security, Programming, Artificial Intelligence, Formal Specification and Verification, Programming Language Concepts, Software Design and Development, Theory of Computing

Taylor's College, Subang Jaya, Malaysia

Sept 2022 - June 2024

A-Levels

- Grades: A*AA (Computer Science, Mathematics, Chemistry)

SMS Sultan Mohamad Jiwa, Kedah, Malaysia

Jan 2017 - March 2022

Malaysian Certificate of Education (SPM)

- Grades: 6A+, 3A
- Award: Petronas Education Sponsorship Programme (PESP) Scholar

RELEVANT PROJECT & COURSEWORK

Pasar Karat MSA | Next.js, TypeScript, Supabase, Tailwind CSS, PostgreSQL

- Architected and deployed a full-stack web marketplace for the Malaysian Student Association to digitize peer-to-peer trading, replacing legacy WhatsApp workflows.
- Engineered secure seller phone authentication, custom session management, and RBAC admin password recovery using SQL migrations and Supabase RLS/Storage policies.
- Built a responsive frontend with full-text search, multi-category filtering, client-side asset uploads, and automated fallback caching to maintain high service availability.

B2B Multi-Tenant Restaurant Platform | Next.js, Node.js, Supabase, WebSockets, PostgreSQL

- Designed a multi-tenant SaaS architecture utilizing Row-Level Security (RLS) for complete cross-tenant data isolation and structured JSON logging.
- Engineered an HMAC/SHA256 signature-verified webhook ingestion pipeline with idempotency tracking to prevent duplicate payment processing.
- Built a real-time Kitchen Display System (KDS) leveraging persistent WebSockets, browser audio alerts, dynamic JSONB menu modifier pricing, and reactive inventory toggling.

Pinecone Healthcare Platform | Java 21, Spring Boot 3, JavaFX, PostgreSQL, Docker

- Developed a full-stack enterprise desktop application with a JavaFX frontend and Spring Boot REST API for encrypted healthcare appointment management and test kit distribution.
- Authored 33+ integration tests using JUnit and MockMvc to ensure compliance and end-to-end reliability across critical medical workflows.
- Containerized backend services with Docker and configured automated CI/CD deployment pipelines on Railway paired with a managed PostgreSQL instance.

Part III Module Advisor | Next.js, TypeScript, Supabase, Vitest, React Hook Form, Zod

- Built a full-stack academic planning tool that parses structured Markdown syllabus data to rank and construct optimal 5-module course paths.
- Developed a multi-factor recommendation engine incorporating user interest weights, dynamic study-time workload classifiers, and dependency-graph prerequisite validation.
- Integrated Zod-validated forms with a server-side Supabase client for persistent user submissions and seamless Vercel deployment.

LEADERSHIP EXPERIENCE

Private Tutor

2022 - 2023

- Delivered tailored academic support in multiple subjects for primary and secondary students, improving examination performance through structured lesson planning and simplified concept delivery.

President, Science and Innovation Club, SMS Sultan Mohamad Jiwa

2020 - 2022

- Led a club of 30+ members in organizing STEM workshops and innovation competitions, fostering technical interest among the student body.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, C, SQL, HTML/CSS, Dart, Haskell

Frameworks & Web: Next.js (App Router), React, Spring Boot, Node.js, JavaFX, Tailwind CSS, Flutter

Cloud, Databases & DevOps: PostgreSQL, Supabase (RLS), WebSockets, Docker, Railway, Vercel, Firebase, MongoDB

Testing & Tools: JUnit, MockMvc, Vitest, React Hook Form, Zod, Git, Unix/Linux, Maven, Gradle

Formal Methods: Rocq (Coq), Dafny, SPIN, Promela

Languages Spoken: English (Fluent), Malay (Native), Thai (Conversational), Mandarin (Conversational)